In [ ]:
```
 EXPERIMENT 5 – DATA VISUALIZATION WITH MATPLOTLIB
PIN NO: 241UDS0101

AIM :-
To visualize and analyze a dataset using Matplotlib plots, including line
charts, scatter plots, subplots, and comparison charts.
```

In [ ]:
```
## Step 1: Import the Required Libraries and Dataset
```

In [3]:
```
import pandas as pd
import matplotlib.pyplot as plt

renewable_241UDS0101 =
pd.read_csv("Experiment_5_241UDS0101_Renewable_Energy.csv")

renewable_241UDS0101.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 7 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   PIN                             10 non-null     object
 1   Year                            10 non-null     int64
 2   Solar_Capacity_GW               10 non-null     float64
 3   Wind_Capacity_GW                10 non-null     float64
 4   Renewable_Generation_TWh        10 non-null     float64
 5   Investment_Billion_USD          10 non-null     float64
 6   Households_Supported_Thousand   10 non-null     int64
dtypes: float64(4), int64(2), object(1)
memory usage: 688.0+ bytes
```

In [ ]:
```
## Step 2: Display the Dataset
```

In [4]:
```
renewable_241UDS0101.head()
```

Out[4]:

	PIN	Year	Solar_Capacity_GW	Wind_Capacity_GW	Renewable_Generation_TWh	Investment_Billion_USD	Households_Supported_Thousand
0	241UDS0101	2016	18.2	9.4	12.5	4.1	1250
1	241UDS0101	2017	20.1	10.8	13.7	4.4	1380
2	241UDS0101	2018	22.6	12.3	15.2	4.7	1495
3	241UDS0101	2019	25.4	14.1	16.8	5.0	1630
4	241UDS0101	2020	28.8	16.7	18.4	5.3	1785

In [5]:
```
renewable_241UDS0101
```

Out[5]:

	PIN	Year	Solar_Capacity_GW	Wind_Capacity_GW	Renewable_Generation_TWh	Investment_Billion_USD	Households_Supported_Thousand
0	241UDS0101	2016	18.2	9.4	12.5	4.1	1250
1	241UDS0101	2017	20.1	10.8	13.7	4.4	1380
2	241UDS0101	2018	22.6	12.3	15.2	4.7	1495
3	241UDS0101	2019	25.4	14.1	16.8	5.0	1630
4	241UDS0101	2020	28.8	16.7	18.4	5.3	1785
5	241UDS0101	2021	32.5	19.2	20.1	5.6	1950
6	241UDS0101	2022	36.9	22.8	22.5	5.9	2140
7	241UDS0101	2023	41.2	26.4	24.7	6.2	2360
8	241UDS0101	2024	46.5	30.7	27.3	6.5	2580
9	241UDS0101	2025	52.1	35.2	30.0	6.8	2810

In [6]:
```
renewable_241UDS0101.columns
```

Out[6]:
```
Index(['PIN', 'Year', 'Solar_Capacity_GW', 'Wind_Capacity_GW',
       'Renewable_Generation_TWh', 'Investment_Billion_USD',
       'Households_Supported_Thousand'],
      dtype='object')
```

In [9]:
```
renewable_241UDS0101["Year"] = pd.to_numeric(renewable_241UDS0101["Year"])
```

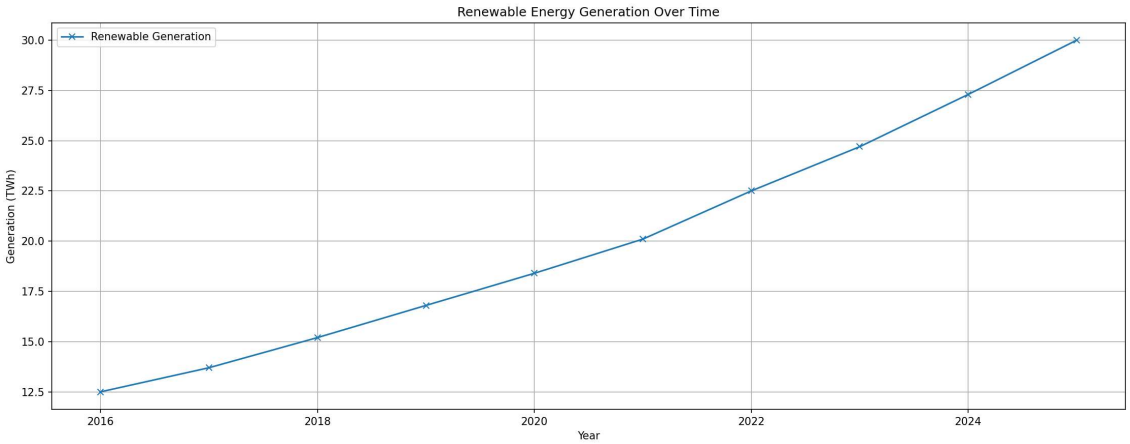
In [ ]:
```
## Step 3: Visualize Data Using Various Plots
```

In [ ]:
```
### 1. Renewable Generation Over Time
```

In [10]:
```
plt.figure(figsize=(15, 6), dpi=150)

plt.plot(
    renewable_241UDS0101["Year"],
    renewable_241UDS0101["Renewable_Generation_TWh"],
    marker="x",
    label="Renewable Generation"
)

plt.title("Renewable Energy Generation Over Time")
plt.xlabel("Year")
plt.ylabel("Generation (TWh)")
plt.tight_layout()
plt.legend()
plt.grid(True)
plt.show()
```



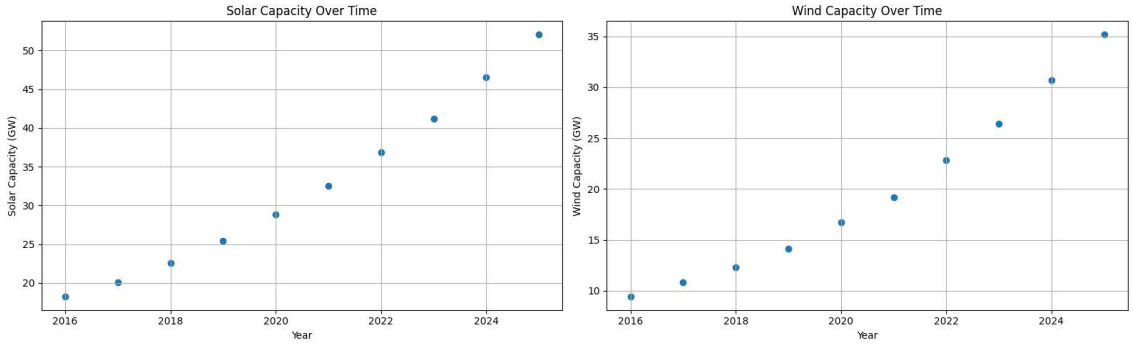
In [ ]:
```
### 2. Subplots: Solar and Wind Capacity
```

In [11]:
```
fig, axes = plt.subplots(1, 2, figsize=(16, 5))

axes[0].scatter(
    renewable_241UDS0101["Year"],
    renewable_241UDS0101["Solar_Capacity_GW"]
)
axes[0].set_title("Solar Capacity Over Time")
axes[0].set_xlabel("Year")
axes[0].set_ylabel("Solar Capacity (GW)")
axes[0].grid(True)

axes[1].scatter(
    renewable_241UDS0101["Year"],
    renewable_241UDS0101["Wind_Capacity_GW"]
)
axes[1].set_title("Wind Capacity Over Time")
axes[1].set_xlabel("Year")
axes[1].set_ylabel("Wind Capacity (GW)")
axes[1].grid(True)

plt.tight_layout()
plt.show()
```



In [ ]:
```
### 3. Comparing Solar Capacity, Wind Capacity, and Renewable Generation
```

In [ ]:
```
Solar and wind capacity are measured in GW, while generation is measured in
TWh. This comparison uses a secondary y-axis so the different units remain
readable.
```

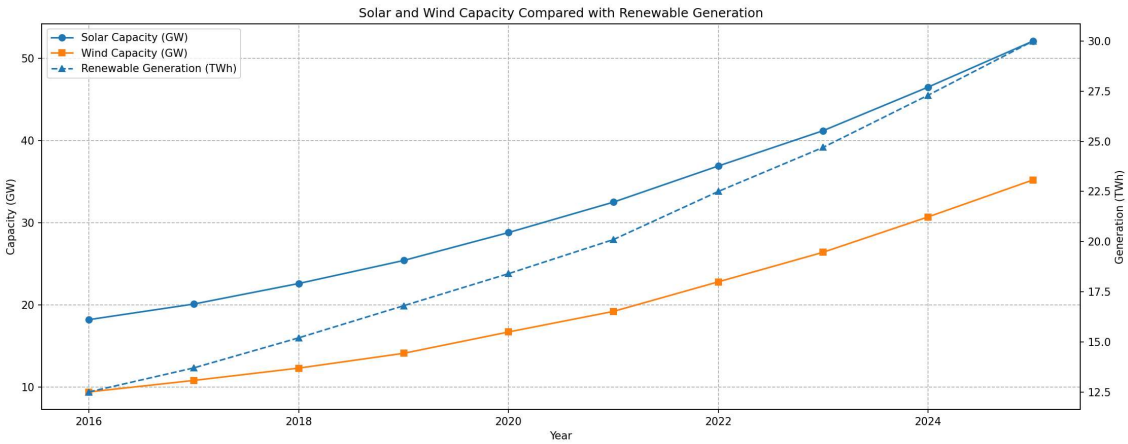
In [12]:
```
fig, ax1 = plt.subplots(figsize=(15, 6), dpi=150)

ax1.plot(
    renewable_241UDS0101["Year"],
    renewable_241UDS0101["Solar_Capacity_GW"],
    marker="o",
    label="Solar Capacity (GW)"
)
ax1.plot(
    renewable_241UDS0101["Year"],
    renewable_241UDS0101["Wind_Capacity_GW"],
    marker="s",
    label="Wind Capacity (GW)"
)
ax1.set_xlabel("Year")
ax1.set_ylabel("Capacity (GW)")
ax1.set_title("Solar and Wind Capacity Compared with Renewable Generation")
ax1.grid(True, linestyle="--")

ax2 = ax1.twinx()
ax2.plot(
    renewable_241UDS0101["Year"],
    renewable_241UDS0101["Renewable_Generation_TWh"],
    marker="^",
    linestyle="--",
    label="Renewable Generation (TWh)"
)
ax2.set_ylabel("Generation (TWh)")

lines1, labels1 = ax1.get_legend_handles_labels()
lines2, labels2 = ax2.get_legend_handles_labels()
ax1.legend(lines1 + lines2, labels1 + labels2, loc="upper left")

plt.tight_layout()
plt.show()
```

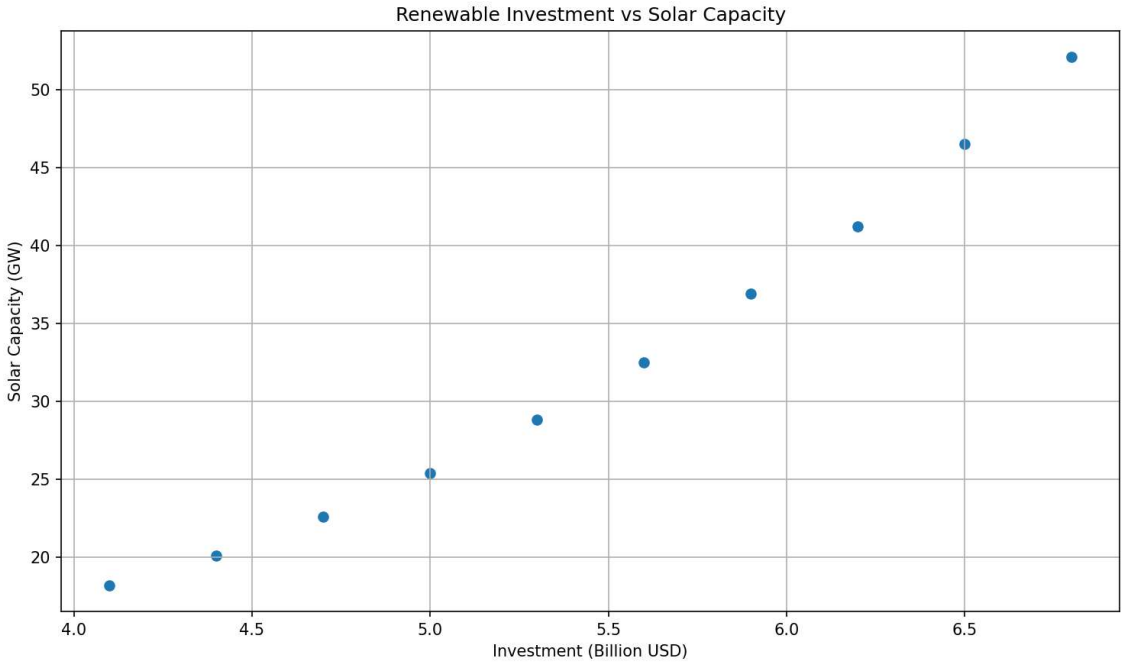


In [ ]:
```
## Step 4: Renewable Investment vs. Solar Capacity
```

In [13]:
```
plt.figure(figsize=(10, 6), dpi=150)

plt.scatter(
    renewable_241UDS0101["Investment_Billion_USD"],
    renewable_241UDS0101["Solar_Capacity_GW"]
)

plt.title("Renewable Investment vs Solar Capacity")
plt.xlabel("Investment (Billion USD)")
plt.ylabel("Solar Capacity (GW)")
plt.grid(True)
plt.tight_layout()
plt.show()
```

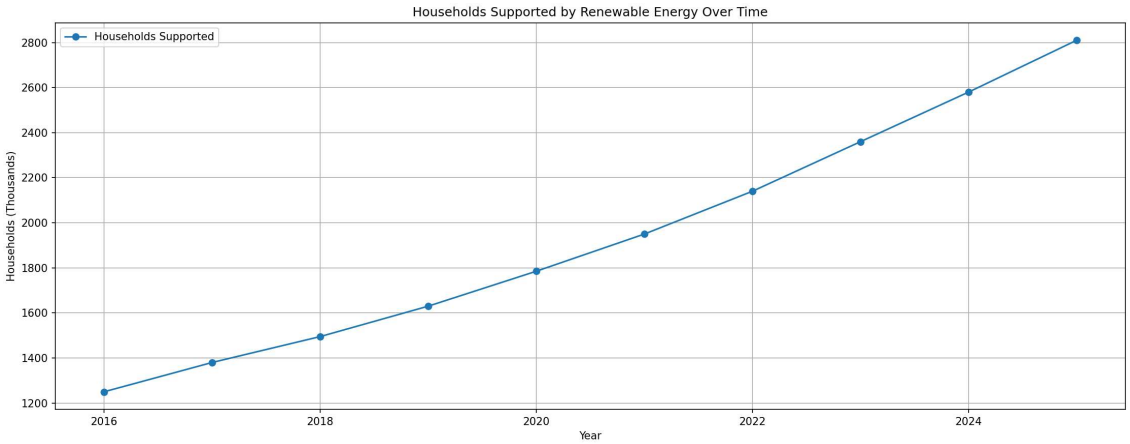


In [ ]:
```
## Step 5: Households Supported Over Time
```

In [14]:
```
plt.figure(figsize=(15, 6), dpi=150)

plt.plot(
    renewable_241UDS0101["Year"],
    renewable_241UDS0101["Households_Supported_Thousand"],
    marker="o",
    label="Households Supported"
)

plt.title("Households Supported by Renewable Energy Over Time")
plt.xlabel("Year")
plt.ylabel("Households (Thousands)")
plt.tight_layout()
plt.legend()
plt.grid(True)
plt.show()
```



In [ ]:
```
RESULT :-
Matplotlib was used to visualize the practice dataset through a line chart,
two scatter subplots, a multi-series comparison chart, an investment-versus-
capacity scatter plot, and a households-supported trend chart.
```